# Mario Feng Wu

+52 (452) 225-0111 | mario.fengw@gmail.com | github.com/MarioFengW | linkedin.com/in/mariofengwu | portfolio/MarioFengW

## EDUCATION

**Tecnológico de Monterrey (ITESM)** August 2023 - June 2027
**Bachelor's of Science in Computer Science and Technology** | Link to all courses Guadalajara, Mexico | GPA: 97/100
**Relevant Coursework:** Development and Deployment of Software Systems (Java), Data Structures and Algorithms (C++) Software Construction and Decision Making (JavaScript), Cybersecurity & Software System (TypeScript).

**National Taiwan University (NTU) | Exchange Program** August 2026 - December 2026
**College of Electrical Engineering and Computer Science (CSIE)** Taipei, Taiwan

## EXPERIENCE

**Research Project | ITESM - Next-Gen Scientist** January 2026 - Ongoing | Hybrid
***AI-Driven Multi-View Cyberattack Detection*** *Python, Scikit-learn, Random Forest, SMOTE*

- Designed a **MultiView Stacking** model with 4 semantic views and a Random Forest meta-learner using 3-fold cross-validation.
- Balanced the **CIC IoT-DIAD 2024** dataset (27.8M+ flows) via undersampling and **SMOTE** inside the stacking folds.
- Beat the single-model baseline in all 20 paired partitions: **F1-Macro 82.56% vs 80.79%**, **Accuracy 90.18%**.
- Ran ablations, LightGBM and stacking controls; preparing a paper for a journal.

**Data Analyst Intern | Amco International** November 2024 - May 2025 | Hybrid
***AI-Driven Automation with ChatGPT*** *Make.com, ChatGPT, Mailchimp*

- Maintained and extended a **Make.com** document pipeline scaled to support **10** events for recurring client campaigns.
- Produced brief-based content with a tuned **ChatGPT** workflow, delivering **10+** assets per event under tight delivery timelines.
- Scheduled **Mailchimp** campaigns for **45,000+** contacts through automated flows across segmented audience list.

**Research Project | ITESM - Rochin MX** September 2024 - November 2024 | Remote
***Real-Time Defect Detection in Fresh Tomatoes Using AI*** *Python, Mask R-CNN, Google Colab*

- Achieved a **90%** tomato-defect detection rate with **Mask R-CNN** and strengthened quality control during grading tests.
- Annotated and augmented **250+** tomato images to improve model training quality for model generalization.

## PROJECTS

**ForgeTask | Development and Deployment of Software Systems** | February 2026 - June 2026
***AI-powered project management and conversational task automation*** *React, Java, OCI, Docker*

- Launched a full-stack platform with **Spring Boot**, **Next.js**, and **JWT authentication** on **Oracle Kubernetes Engine**.
- Built a **RAG** reporting pipeline over **Oracle Autonomous Database** vector search, automating executive KPI reports.
- Reached **71%** sprint task completion while cutting actual hours **35%** below estimates across 4 sprints.
- Automated **blue-green CI/CD** with **OCI DevOps**, **Docker**, and **Kubernetes**, backed by **Selenium** end-to-end testing.

**In & Out | Cybersecurity & Software System** | August 2025 - December 2025
***Digital transformation for secure and efficient donation logistics*** *React Native, TypeScript, Supabase*

- Developed for **Banco de Alimentos Jalisco** with **React Native** and **Supabase**, serving donors, transport operators, and admins.
- Established a **7-state** project lifecycle with role-scoped action cards, persisting real-time transitions via **Supabase** Realtime.
- Implemented automatic vehicle-assignment logic matching availability, weight, and load type on project acceptance.

**Connect the Schools | Software Construction** | February 2025 - June 2025
***Educational Support Platform Connecting Schools with Partners*** *React, Node.js, PostgreSQL*

- Created a scalable web platform for **3,000+** schools to Mexicanos Primero Jalisco, helping bridge educational resource gaps.
- Delivered a comprehensive needs assessment system with similarity-based matching for **8+** categories of educational support.

## VOLUNTEERING

**PawnGuard** | *Board of Directors* | September 2025 - June 2026

- Contributed UX design ideas and co-developed the frontend of the Hack2Dawn CTF website for 70 participants.
- Shared knowledge, resources, and challenges related to cybersecurity to foster collaborative learning and technical growth.

**Guadalahacks & TECoding** | *Board of Directors* June 2024 - December 2025

- Organized hackathon preparation sessions on project ideation, team formation, and coding practices for student teams.

## SKILLS & CERTIFICATIONS

**Certifications:** NextGen University Program for Cloud&DevOps | Link to all certifications
**Languages:** Python, C++, SQL, Java, JavaScript, HTML/CSS, MATLAB, Bash, TypeScript.
**Technologies:** React, React Native, Node.js, Vue.js, Flask, Spring Boot, Next.js, Scikit-learn, NumPy, Pandas, PostgreSQL, MariaDB, Supabase, Oracle Autonomous Database, Kubernetes.
**Developer Tools:** Git, GitHub, VS Code, Google Colab, Docker, OCI DevOps, Selenium, Oracle Kubernetes Engine.